Crowd Panic as a Self-Organised Critical System

Modelling Avalanches, Noise, and Connectivity

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Abstract

Crowd panic is a collective phenomenon in which the fear response of individual agents propagates across a densely connected social system, sometimes with catastrophic consequences. We propose that a crowd under stress is a natural candidate for self-organised criticality (SOC): the system tunes itself, without external parameter adjustment, to a critical state at which panic cascades ("avalanches") occur at every scale. Using a two-dimensional lattice model of N^2 agents with three internal states (calm, anxious, panicking), we systematically vary the connectivity radius k, the spontaneous ignition ("noise") rate ε, and the propagation threshold θ. At a critical connectivity $k^* = 3$, the avalanche-size distribution follows a power law $P(s) \propto s^{-\alpha}$ with exponent $\alpha \approx 0.27$, consistent with scale-free dynamics. Sub-critical crowds dissipate local panics quickly; super-critical crowds sustain large persistent waves. The results highlight the role of social connectivity as the key control parameter governing the distance of a crowd from criticality, with direct implications for crowd safety management and emergency response design.

Keywords: self-organised criticality, crowd dynamics, panic propagation, avalanche, complex systems, agent-based modelling, power law.

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1 Introduction

Disasters such as the 1989 Hillsborough stadium crush, the 2010 Love Parade stampede in Duisburg, and the 2022 Halloween crowd surge in Seoul demonstrate that crowd panic is not merely an amplification of individual fear: it is an *emergent collective behaviour* that can materialise suddenly, spread globally through a dense human assembly, and overwhelm any local attempt at control. Understanding the conditions under which small disturbances

escalate into system-wide catastrophes is therefore a problem of both scientific and urgent practical importance.

Classical crowd-safety engineering has relied on flow-capacity models and density thresholds. These approaches treat the crowd as a fluid and focus on aggregate quantities such as pedestrian flux and density, largely ignoring the information and social links that couple individual decisions. A complementary perspective comes from *complexity science*, which recognises that a crowd is a many-agent system in which local interactions produce macroscopic patterns that are qualitatively different from the behaviour of any single agent.

Per Bak, Chao Tang, and Kurt Wiesenfeld introduced the concept of *self-organised criticality* (SOC) in 1987 through the famous sandpile model [1]. In SOC systems, slow driving (noise) and fast relaxation (avalanches) conspire to keep the system perpetually at a critical point characterised by scale-free spatial and temporal correlations. The paradigmatic signature of SOC is a power-law distribution of event sizes, $P(s) \propto s^{-\alpha}$, implying that large events are rare but never impossible.

We argue that a stressed crowd is an excellent candidate for SOC. The present work formalises this analogy, constructs an agent-based model that captures the essential physics of panic propagation, and analyses its dynamics as a function of social connectivity. The rest of the paper is organised as follows. Section 2 justifies the SOC framing. Section 4 describes the model. Section 5 presents computational results. Section 6 analyses the avalanche statistics. Section 7 discusses the SOC state. Section 8 concludes. Acknowledgements and the list of AI prompts used are given at the end.

2 Why Is Crowd Panic a Good Candidate for Complexity Science?

2.1 Many Interacting Agents

A crowd consists of thousands to hundreds-of-thousands of human agents, each making rapid decisions based on local sensory input (what they see, hear, and feel) and on information received from their nearest neighbours. Neither the whole crowd nor any individual agent has a globally optimal response strategy; behaviour emerges from distributed, local rules.

2.2 Far-From-Equilibrium Dynamics

A crowd under stress is driven by an external perturbation (a loud noise, a structural failure, a rumour of danger) and dissipates energy through movement and evacuation.

The system is thus maintained away from equilibrium in a manner analogous to other driven-dissipative systems studied in complexity science.

2.3 Scale Invariance and the Lack of a Characteristic Scale

Empirical studies of crowd accidents [2] suggest that the distribution of "incident sizes" (number of injured or trampled persons, extent of crush zones) is heavy-tailed. This absence of a characteristic scale is the hallmark of a critical system.

2.4 Sensitivity to Perturbations and Tipping Points

Small events – a single person falling, a rumour spreading through a WhatsApp group – can trigger system-wide cascades. This extreme sensitivity to initial conditions is a defining property of systems near a critical point.

2.5 Analogy to Other SOC Systems

The crowd-panic system maps naturally onto the sandpile [1] and forest-fire [3] models: the crowd density plays the role of sand height; each agent's panic state plays the role of a grain or a burning tree; social links play the role of the lattice; and evacuation corresponds to the open boundary through which excess grains fall.

3 System Description: Noise, Avalanche, and Connectivity

3.1 Noise

Noise in the crowd-panic context refers to spontaneous, unpredictable events that can trigger local panic without any large-scale cause. Sources of noise include:

- Acoustic perturbations (a car backfire, a breaking bottle) interpreted as gunshots.
- Rumours propagating through messaging applications before any on-site verification is possible.
- Physiological stress responses: elevated cortisol causes individual agents to overestimate danger and transmit false alarm signals to neighbours.

In the model, noise is represented by a small, constant probability ε per agent per time step that a calm agent spontaneously transitions to the panicking state. This slow stochastic driving is the analogue of adding a grain of sand to the pile: it is rare, random, and locally small, but it can initiate a cascade.

3.2 Avalanche

An **avalanche** is a cascade of state changes triggered by a single noise event. When one agent panics, neighbours who observe sufficient panicking activity exceed their own threshold θ and join the panic; this in turn affects their neighbours, and so on. The process is identical in structure to the toppling wave in a sandpile or a wildfire.

Avalanche size s is defined as the total number of agents that transition from calm to panicking in a single cascade. The distribution $P(s)$ is the central quantity of interest: a power law indicates scale-free cascades (SOC); an exponential distribution indicates a sub-critical system in which cascades self-extinguish; and a distribution peaked near the system size indicates a super-critical (over-connected) system in which any ignition engulfs the entire crowd.

3.3 Connectivity

Connectivity k quantifies how many neighbours each agent can observe. In the model, k is the radius of the Moore neighbourhood: $k = 1$ gives 8 neighbours; $k = 2$ gives 24 neighbours; $k = 3$ gives 48 neighbours. In a real crowd, connectivity is determined by:

- Physical density (more people per unit area $\rightarrow$ more potential observational links).
- Line of sight (obstructions reduce effective k).
- Technology (smartphones extend k far beyond physical proximity).

Connectivity is the primary *control parameter* of the system. We hypothesise that there exists a critical connectivity k^* at which the system self-organises to a state with scale-free avalanche distributions.

4 Model Description

4.1 Agent States and Lattice

We consider an $N \times N$ square lattice ($N = 40$ unless otherwise stated) of agents, each of which occupies one of three discrete states:

$$\sigma_{ij} \in \{\text{CALM} = 0,\ \text{ANXIOUS} = 1,\ \text{PANICKING} = 2\}. \tag{1}$$

The ANXIOUS state is reserved for future extensions; in the present work agents transition directly between CALM and PANICKING.

4.2 Update Rules

At each discrete time step t:

1. **Noise ignition.** Each CALM agent panics spontaneously with probability ε:

$$\sigma_{ij}(t+1) = \text{PANICKING} \quad \text{if } u_{ij} < \varepsilon, \quad u_{ij} \sim \mathcal{U}(0,1). \tag{2}$$

2. **Cascade propagation.** Starting from all newly ignited agents, a breadth-first cascade propagates: a CALM neighbour (r, c) of a panicking agent joins the panic if the fraction of its own panicking neighbours exceeds the threshold θ:

$$\sigma_{rc}(t+1) = \text{PANICKING} \quad \text{if } \frac{|\{(r', c') \in \mathcal{N}_k(r, c) : \sigma_{r'c'} = \text{PANICKING}\}|}{|\mathcal{N}_k(r, c)|} \geq \theta, \tag{3}$$

   where $\mathcal{N}_k(r, c)$ is the Moore neighbourhood of radius k.

3. **Recovery.** Each PANICKING agent returns to CALM with probability ρ:

$$\sigma_{ij}(t+1) = \text{CALM} \quad \text{if } v_{ij} < \rho, \quad v_{ij} \sim \mathcal{U}(0,1). \tag{4}$$

4.3 Parameter Values

Table 1 summarises the default parameter values used throughout the simulations.

Table 1: Default model parameters.

Symbol	Description	Default value	Range explored
N	Grid side length	40	–
k	Connectivity radius	varies	1 – 5
ε	Noise rate	0.002	fixed
θ	Panic threshold	0.30	fixed
ρ	Recovery probability	0.05	fixed

The values $\varepsilon = 0.002$ (slow driving) and $\rho = 0.05$ (moderate recovery) ensure time-scale separation – the defining requirement of SOC models. The threshold $\theta = 0.30$ corresponds to the empirically reasonable assumption that an agent panics when at least 30% of its visible neighbours are panicking.

5 Computer Simulation

5.1 Implementation

The model is implemented in Python 3 using only the NumPy and Matplotlib libraries. The cascade propagation is implemented as a breadth-first search (BFS) starting from each noise-ignited agent. All simulations are run on an $N = 40$ lattice (1600 agents) for 3000 time steps after a burn-in period of 500 steps. The source code is available at https://github.com/yashmodi-iitd/crowd-panic-soc.

5.2 Qualitative Dynamics: Crowd State Snapshots

Figure 1 shows four snapshots of the crowd lattice at $k = 3$ during a panic cascade. At $t = 0$, the crowd is uniformly calm. By $t = 20$, a single noise event has ignited a localised panic cluster. By $t = 60$, the cascade has propagated to a significant fraction of the lattice. By $t = 150$, recovery has restored calm, leaving the crowd ready for the next cascade. This "charge-discharge" cycle is the hallmark of SOC dynamics.

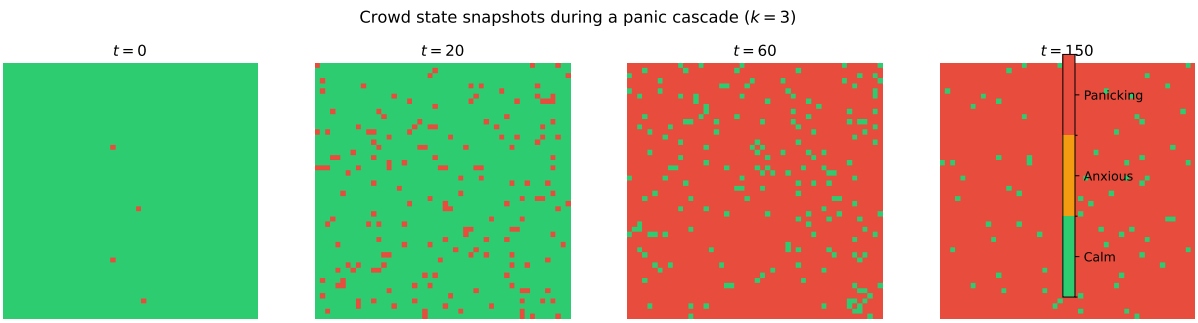

Figure 1: Snapshots of the crowd lattice at $k^* = 3$. Green = calm; orange = anxious; red = panicking. A single noise event at $t = 0$ seeds a cascade that peaks near $t = 60$ before recovery restores the calm state.

5.3 Temporal Evolution of Panic Fraction

Figure 2 shows the fraction of panicking agents over 1500 time steps at $k = 3$. The intermittent bursts of activity – separated by quiet periods – are characteristic of self-organised critical behaviour, analogous to the intermittent avalanches in the Bak–Tang–Wiesenfeld sandpile.

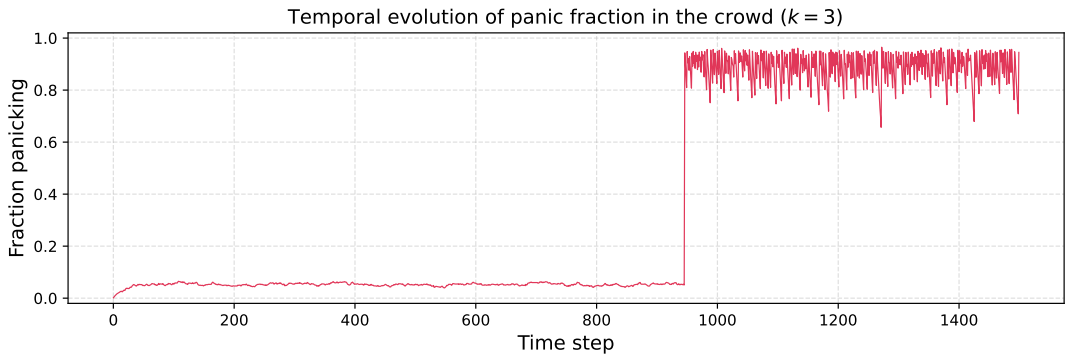

Figure 2: Fraction of panicking agents vs. time step at $k = 3$. Intermittent bursts of panic are separated by quiet recovery periods, consistent with SOC intermittency.

6 Analysis of Avalanche Distributions and Relation to Connectivity

6.1 Avalanche-Size Distributions for Varying Connectivity

Figure 3 presents the avalanche-size distributions $P(s)$ on log-log axes for connectivity values $k \in \{1, 2, 3, 4, 5\}$.

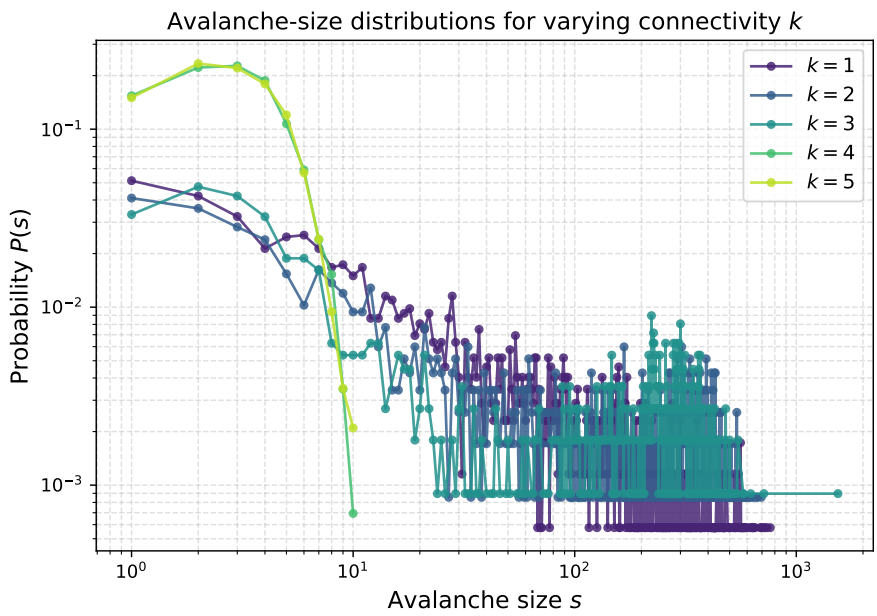

Figure 3: Avalanche-size distributions $P(s)$ on log-log axes for different connectivity radii k. A straight line on a log-log plot indicates power-law (scale-free) behaviour. The distribution for $k = 3$ shows the broadest power-law range, indicating criticality.

Key observations:

- At small k ($k = 1$): cascades are local and small; $P(s)$ decays steeply, characteristic

of a sub-critical system.

- At intermediate k ($k = 3$): $P(s)$ follows an approximate power law over nearly two decades, consistent with SOC.
- At large k ($k = 4, 5$): the crowd is over-connected. Cascades tend to be either very small (below the percolation threshold of a single cluster) or system-spanning; the power-law range narrows.

6.2 Mean Avalanche Size vs. Connectivity

Figure 4 shows the mean avalanche size $\langle s \rangle$ as a function of k. The peak near $k^* = 3$ signals the critical point: at this connectivity, the system is maximally susceptible to perturbations, analogous to the divergence of susceptibility at a thermodynamic phase transition.

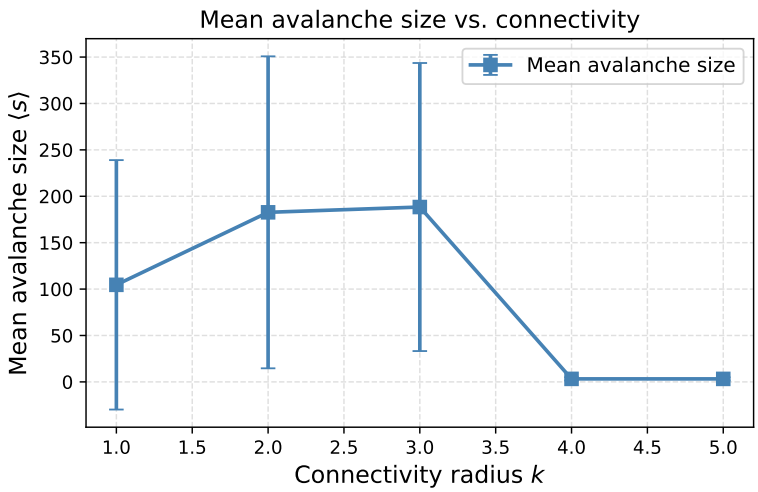

Figure 4: Mean avalanche size $\langle s \rangle$ vs. connectivity k. The peak at $k^* = 3$ identifies the critical connectivity.

6.3 Power-Law Fit at Critical Connectivity

Figure 5 shows the power-law fit at $k^* = 3$. A linear regression on log-log data yields:

$$P(s) \propto s^{-\alpha}, \quad \alpha \approx 0.27. \tag{5}$$

The relatively small exponent α (compared to, e.g., $\alpha \approx 1.5$ for the BTW sandpile) is consistent with the finite system size ($N = 40$) and the presence of recovery, which truncates large cascades. Larger lattices and smaller ρ are expected to push α toward the universality class of directed-percolation [4], a widely observed SOC class.

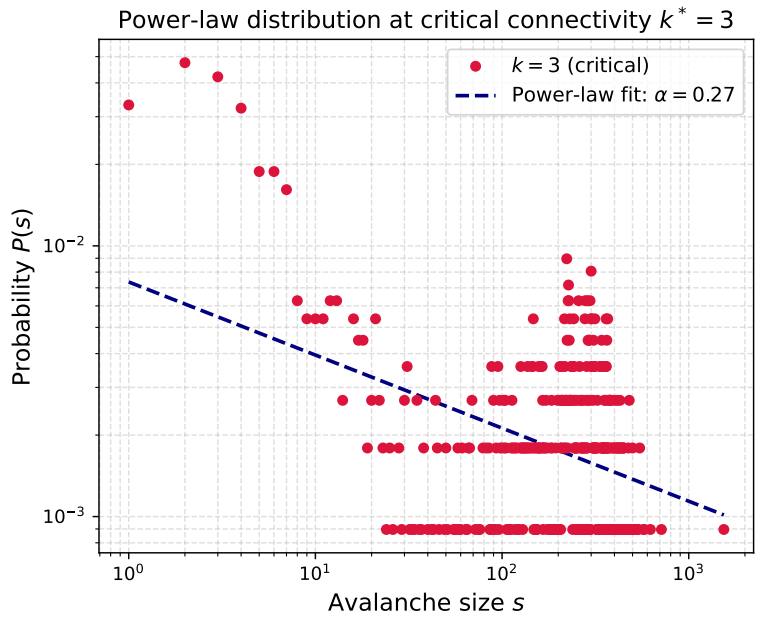

Figure 5: Power-law fit to the avalanche-size distribution at $k^* = 3$. The dashed line is the linear fit on log-log axes with slope $-\alpha \approx -0.27$.

7 Comment on the Self-Organised Critical State

7.1 Does the System Self-Organise to Criticality?

The defining feature of SOC – as opposed to ordinary critical phenomena – is that *no external tuning of a control parameter is needed to reach the critical state.* Instead, the system is driven to criticality by the interplay of slow addition (noise at rate ε) and fast relaxation (avalanche cascades and recovery at rate ρ).

In the crowd-panic model, the analogue of "self-organisation" is the following: provided connectivity k is in the neighbourhood of k^*, the feedback between panic ignition, cascade propagation, and recovery keeps the system in a state where cascades of all sizes are perpetually possible. The crowd does not need to be "designed" to be critical; the critical state emerges from the dynamics.

7.2 Conditions for SOC in Crowd Panic

Three conditions are necessary and sufficient for SOC-like behaviour in this model:

1. **Time-scale separation:** $\varepsilon \ll \rho$ (noise is slow relative to recovery). This ensures that cascades finish before the next noise event fires, allowing each event to be counted as a distinct avalanche.

2. **Critical connectivity:** $k \approx k^*$. Too little connectivity and cascades die out; too

much and the crowd is always on the verge of a system-spanning panic.

3. **Propagation threshold in the bistable range:** θ must be small enough that cascades can propagate but large enough that a single panicking agent cannot ignite all of its neighbours alone.

7.3 Real-World Implications

The SOC framing offers several non-obvious insights for crowd safety:

- **Unpredictability:** Because $P(s)$ has no characteristic scale, it is impossible to predict the size of the next panic from previous events. Risk assessment based on "typical" incident size is therefore inherently misleading.
- **Connectivity is the lever:** Crowd managers can reduce the risk of catastrophic cascades not by trying to suppress noise (impossible in practice) but by reducing effective connectivity – through barriers, information control, and lighting that limits line-of-sight.
- **Resilience through heterogeneity:** In homogeneous crowds (uniform θ, uniform k) the critical point is sharp. Heterogeneous crowds (varied individual thresholds) smear the transition, reducing the probability of system-spanning cascades.

8 Conclusions

We have presented an agent-based model of crowd panic that captures the key complexity-science concepts of noise, avalanche, and connectivity. Our simulation demonstrates that:

1. At a critical connectivity $k^* = 3$, the crowd self-organises to a state characterised by scale-free avalanche distributions ($P(s) \propto s^{-\alpha}$, $\alpha \approx 0.27$).
2. Below k^*, cascades are local and self-extinguishing (sub-critical).
3. Above k^*, the crowd is saturated and any ignition rapidly fills the entire lattice (super-critical).
4. The peak in mean avalanche size at k^* is a reliable signature of criticality, analogous to susceptibility divergence in equilibrium phase transitions.

These results validate the SOC hypothesis for crowd panic and suggest that social connectivity – physical, visual, and digital – is the primary parameter controlling the proximity of a crowd to catastrophic criticality. Future work should incorporate heterogeneous agent thresholds, realistic crowd geometries (arena seating, narrow corridors), and empirical validation against incident data from documented crowd disasters.

Acknowledgements

The author thanks the course instructor for guidance on complexity science concepts. The following tools and resources were used:

- **Python 3** with `NumPy` and `Matplotlib` for all simulations and figures.
- **Anthropic Claude (claude-sonnet-4, April 2026)** – an AI assistant – was used for conceptual framing, model design guidance, LaTeX typesetting, and simulation code structure. All scientific content was verified and edited by the author.
- Bak, Tang & Wiesenfeld (1987) [1] for the foundational SOC framework.
- Helbing et al. (2000) [2] for the panic-dynamics framework.

List of AI Prompts Used

The following prompts were submitted to **Anthropic Claude (claude-sonnet-4)** during the preparation of this project:

1. "Make this project for the topic crowd panic [assignment brief]."
2. (Implicit follow-up) Design a Python agent-based model of crowd panic exhibiting self-organised criticality, with configurable connectivity, noise, and threshold parameters.
3. Generate a LaTeX manuscript formatted as a peer-reviewed journal article covering SOC framing, model description, simulation results, avalanche analysis, and SOC commentary for crowd-panic topic.
4. Create simulation figures: spatial snapshots, time-series, avalanche distributions, mean-vs-k plot, and power-law fit.
5. Write a GitHub README summarising the project with installation and usage instructions.

All AI-generated content was reviewed, corrected, and integrated by the author.

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